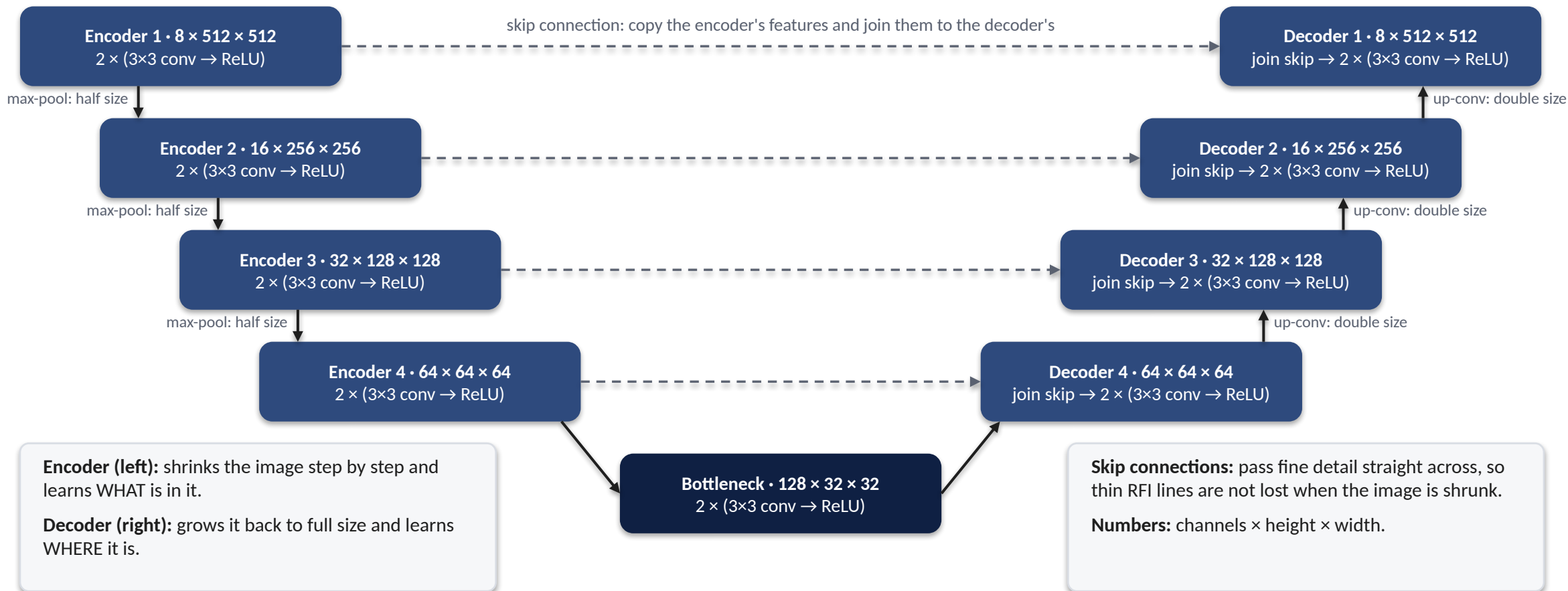


➤ Starting point: a basic U-Net

▼ Input: spectrogram, $1 \times 512 \times 512$ (time $\times$ frequency)

Output: an RFI probability for every pixel ▲

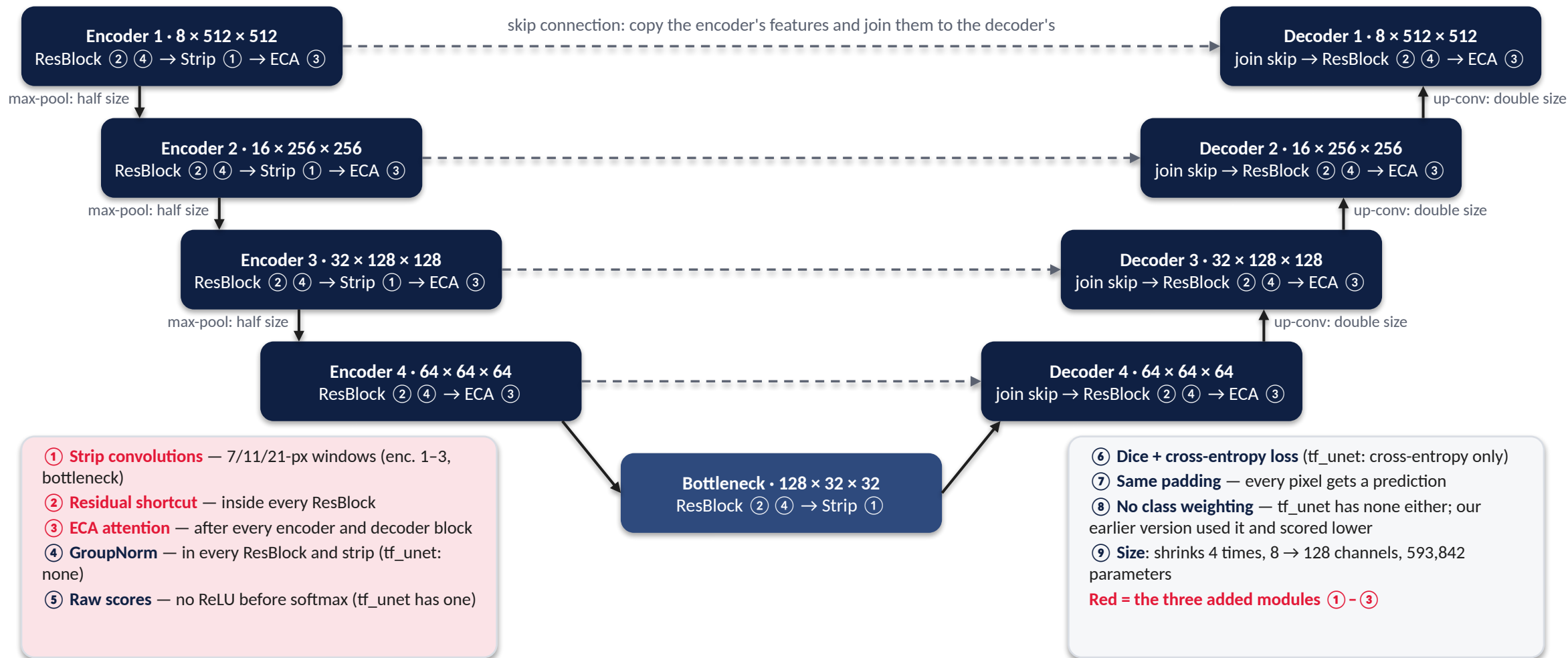


Drawn at our model's sizes so slides 1 and 2 line up box for box. Akeret's tf_unet (our baseline) is this design, but shrinks only twice ($32 \rightarrow 64 \rightarrow 128$ channels, 465,986 parameters) and trims the border, so 512×512 in gives 472×472 out.

➤ Our model: the same U, changed inside

▼ Input: spectrogram, $1 \times 512 \times 512$ — no border trimming ⑦

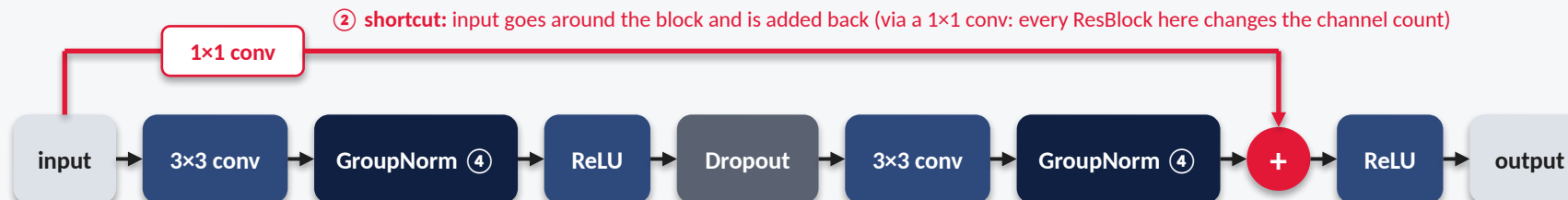
1×1 conv $\rightarrow$ 2 raw scores $\rightarrow$ softmax ⑤ $\rightarrow$ RFI probability ▲



➤ Inside one level, step by step

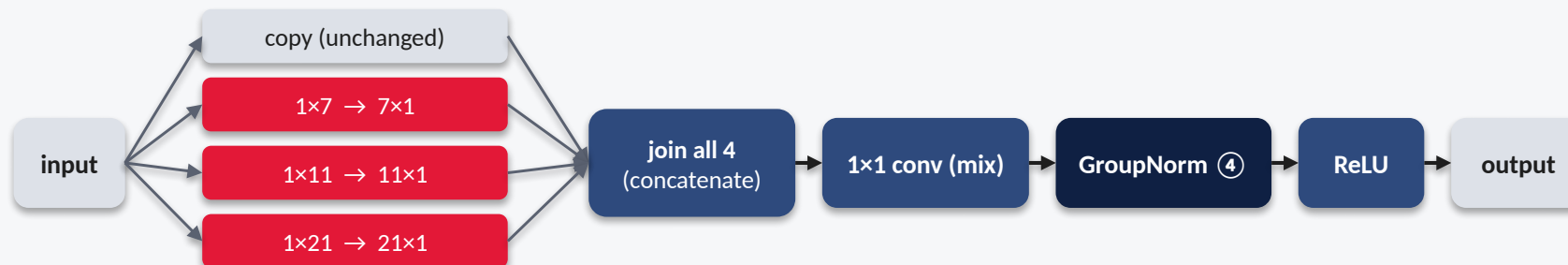
ResBlock ② ④

Two 3×3 convolutions, each tidied by GroupNorm; the input is added back at the end. Dropout works in training only.



Strip module ①

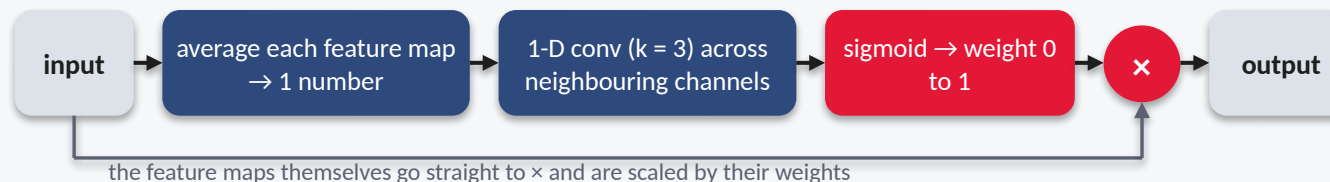
Looks at wider areas (7, 11, 21 px) cheaply: a thin filter along frequency, then one along time.



Each path = 1×K along frequency, then K×1 along time, each channel on its own → together it covers a K×K square (measured)

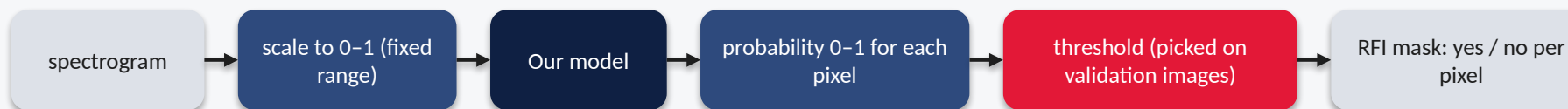
ECA attention ③

Gives each feature map a weight between 0 and 1, then scales it.



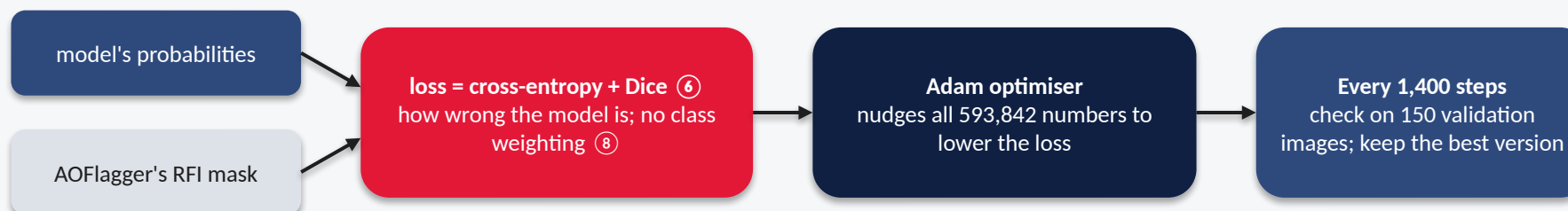
➤ From output to RFI mask, and how it learns

Using the model
after training



Training

28,000 steps, one image per step



How it is scored: 109 test images labelled by a human expert; F1 at the best threshold (the standard for this benchmark); 3 runs each, mean $\pm$ spread.

Our model 0.6603 ± 0.0040 vs Akeret's `tf_unet` $0.5482 \pm 0.0139 \rightarrow$ about +0.11.

Which parts matter? Removing ① ② ③ gives 0.6585 — not significant ($p = 0.64$). Removing ④ as well gives 0.6495 — also not significant ($p = 0.48$).

So the gain over `tf_unet` sits mainly in ⑤ ⑥ ⑦ ⑨ (raw output, Dice loss, same padding, size and training setup) — not yet tested one at a time.